



मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान भोपाल, भोपाल – 462003

Maulana Azad National Institute of Technology Bhopal, Bhopal – 462003

(An Institution of National Importance under Ministry of Education, Govt. of India)

Department of Computer Science & Engineering

B.Tech. V Semester (EE1)

Data Structures (CSE352)

Assignment-II

1. Suppose, an array $A[-15 \dots 64]$ is stored in a memory whose starting address is 459. Assume that the word size for each element is 2. Then obtain the following:
 - a) How many elements are there in the array?
 - b) If one word of the memory is equal to 2 bytes, then how much memory is required to store the entire array?
 - c) What is the location for $A[50]$?
2. Consider the base address of an array $A[1300 \dots 1900]$ as 1020 and the size of each element is 4 bytes in the memory. Find the address of $A[1700]$.
3. Consider an array, $\text{arr}[1 \dots 10][1 \dots 15]$ with base value 1000 and the size of each element is 4 Byte in memory. Find the address of $\text{arr}[8][6]$ when elements are stored as row-major order.
4. Given an array, $\text{arr}[2 \dots 12][3 \dots 18]$ with a base address of 200 and each element having a size of 2 bytes in memory, find the address of $\text{arr}[10][5]$ using column-major order.
5. Given an array, $\text{arr}[1:9, -4:1, 5:10]$ with a base value of 400 and the size of each element is 2 Bytes in memory. find the address of element $\text{arr}[5][-1][8]$ using row-major order.
6. Consider 3-dimensional Array $A[90][30][40]$ stored in a linear array in column-major order. If the base address starts at 10. The location of $A[20][20][30]$ is _____.
[Assume the first element is stored at $A[1][1][1]$ and each element takes 1 B].